Your submitted solution is only **valid if it contains the following paragraphs** at the top of the program. Please ***fill*** in ***your name*** and ***Neptun code***.

```
/*<NAME>, <NEPTUN>
This solution was prepared and submitted by the student stated above for the assignment
of the Programming course. I declare that this solution is my own work. I have not
copied or used third party solutions. I have not passed my solution to my classmates,
neither made it public.
Students’ regulation of Eötvös Loránd University (ELTE Regulations Vol. II. 74/C.§)
states that as long as a student presents another student’s work - or at least the
significant part of it - as his/her own performance, it will count as a disciplinary
fault. The most serious consequence of a disciplinary fault can be dismissal of the
student from the University.*/
```

When submitting on TMS upload the program with the output enabled that only prints the result of the specific subtask e.g.: *if you are solving a) do not include the outputs for tasks b), c), d)***!**

Galaxy Explorers

The galaxy is bustling with spacefarers collecting resources, building alliances, and engaging in epic missions. The Galactic Council has started tracking their activities to understand how these explorers contribute to the galaxy's prosperity. Each record includes details about the explorer's actions, like the resource they gathered, its quantity, the energy consumed during the task and the mission’s danger level.

Input

The first line of input contains N (1 ≤ N ≤ 50), the number of recorded missions, and T (1 ≤ T ≤ 100), a threshold for tasks (b) and (d). Each of the next N lines contains a string E the explorer's name (e.g., Luke, Leia, Han), string R the resource gathered (e.g., Kyber, Spice, Bacta), an integer Q the quantity of the resource collected (1 ≤ Q ≤ 100), a floating-point number C for consumed energy during the mission (0.1 ≤ C ≤ 100.0) and an integer D for the mission’s danger level (1 ≤ D ≤ 5) .

Write a program that solves the following tasks:

Tasks:

a) Print the number of missions where an explorer consumed less than 30.0 units of energy.

b) Count the number of records which gathered resource value is lower than to the given threshold. Output this count as a single number.

c) For each unique resource, calculate the total quantity collected and print the resource name along with its total. Maintain the order in which the resources first appear.

d) Find the number of the first record for a specific explorer named Luke if the summation of their gathered resource quantities exceeds the given threshold T and the associated mission danger levels are all greater or equal to 3.

Output

Each task's output should be separated by new lines. Do *NOT* print the text in red; the red color is used here solely to illustrate and distinguish each subtask's output for clarity.

Example

Input

```
4 20
Leia Kyber 15 45.5 3
Luke Spice 25 71.2 4
Han Bacta 10 21.7 1
Leia Kyber 20 36.8 2
```

Output

```
Task a):
1

Task b):
2

Task c):
Kyber 35
Spice 25
Bacta 10

Task d):
Luke 25
```